Random Variables

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1 Setting

Fix a probability space $(\Omega, \mathcal{F}, P)$ and write $\mathcal{B}(\mathbb{R})$ for the Borel σ-algebra of $\mathbb{R}$ (the smallest σ-algebra containing the open intervals).

Definition 1 (Random variable). *A function $X : \Omega \to \mathbb{R}$ is a* random variable *if*

$$X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\} \ \in \ \mathcal{F} \qquad \text{for every } B \in \mathcal{B}(\mathbb{R}).$$

Remark 2. *Equivalently, X is a random variable iff $\{X \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. This follows because $\{(-\infty, x] : x \in \mathbb{R}\}$ generates $\mathcal{B}(\mathbb{R})$ and X^{-1} commutes with countable unions, intersections, and complements.*

Definition 3 (Distribution function). *The* cumulative distribution function *(CDF) of a random variable X is*

$$F_X : \mathbb{R} \to [0,1], \qquad F_X(x) := P(X \leq x) = P\big(X^{-1}((-\infty, x])\big).$$

The function F_X is non-decreasing, right-continuous, with $F_X(-\infty) = 0$ and $F_X(+\infty) = 1$.

2 Algebra of random variables

We now prove that the sum of two random variables is again a random variable. This is the *prototype* for showing that products, maxima, minima, and limits of random variables remain random variables.

Lemma 4 (Pre-image of an open set). *Let $X : \Omega \to \mathbb{R}$ satisfy $\{X \leq x\} \in \mathcal{F}$ for all $x \in \mathbb{R}$. Then for every open set $U \subseteq \mathbb{R}$, $X^{-1}(U) \in \mathcal{F}$.*

Proof. Every open subset of $\mathbb{R}$ is a countable union of open intervals. For an interval (a, b),

$$X^{-1}((a,b)) = \{X < b\} \cap \{X > a\} = \Big(\bigcup_{n=1}^{\infty} \{X \leq b - 1/n\}\Big) \cap \{X \leq a\}^c,$$

which is a countable Boolean combination of sets in $\mathcal{F}$, hence in $\mathcal{F}$. Taking countable unions preserves this. □

Theorem 5 (Sum of random variables is a random variable). *If $X, Y : \Omega \to \mathbb{R}$ are random variables, then $Z := X + Y$ is a random variable.*

Proof. By the remark, it suffices to show $\{Z \le c\} \in \mathcal{F}$ for every $c \in \mathbb{R}$. Equivalently, since $\{Z \le c\} = \{Z < c\}^c \cup \{Z = c\}$ and complements stay in $\mathcal{F}$, we instead exhibit $\{Z < c\}$ as a measurable set — this is the cleaner form and uses Lemma 4 indirectly.

Observe that $X(\omega) + Y(\omega) < c$ holds iff there exists a rational q with

$$X(\omega) < q \quad \text{and} \quad Y(\omega) < c - q.$$

($\Rightarrow$): If $X(\omega) + Y(\omega) < c$, set $\varepsilon = c - X(\omega) - Y(\omega) > 0$; by density of $\mathbb{Q}$ in $\mathbb{R}$, pick $q \in \mathbb{Q}$ with $X(\omega) < q < X(\omega) + \varepsilon/2$. Then $q < X(\omega) + \varepsilon/2$ gives $c - q > c - X(\omega) - \varepsilon/2 = Y(\omega) + \varepsilon/2 > Y(\omega)$. ($\Leftarrow$): If $X(\omega) < q$ and $Y(\omega) < c - q$, add the inequalities.

Hence

$$\{Z < c\} \;=\; \bigcup_{q \in \mathbb{Q}} \Big(\{X < q\} \cap \{Y < c - q\} \Big).$$

The collection $\mathbb{Q}$ is countable. Each $\{X < q\} = X^{-1}((-\infty, q))$ and each $\{Y < c - q\} = Y^{-1}((-\infty, c - q))$ is the pre-image of an open set, hence in $\mathcal{F}$ by Lemma 4. A countable union of intersections of $\mathcal{F}$-sets is in $\mathcal{F}$, so $\{Z < c\} \in \mathcal{F}$. Finally $\{Z \le c\} = \bigcap_{n=1}^{\infty} \{Z < c + 1/n\} \in \mathcal{F}$, proving the theorem. $\square$

Remark 6. *The same density-of-rationals trick proves* $XY, \max(X, Y), \min(X, Y)$ *are random variables. Iterating gives that every polynomial in finitely many random variables is a random variable, and combining with Theorem 3 of the README extends the result to limits.*